

Chuyue Zhang

Software Engineer

📍 Toronto, ON, Canada ✉ zhangchuyue.zhang@mail.utoronto.com ☎ 14373438066

in Chuyue Zhang 🍲 IronDumpling

Skills

Languages: C++, Modern C++ (11/14/17), C#, Java, Go, Python, Rust, SQL

Databases & Data Platforms: PostgreSQL, MySQL, MongoDB, Redis, Cassandra, DynamoDB, ClickHouse; Query Optimization, Partitioning & Indexing Strategies, Execution Planning; Data Modeling (OLTP/OLAP), Columnar Storage, Materialized Views, Parquet; Data Warehousing (Snowflake), PySpark

Web & Infrastructure: React.js, Node.js, Express.js, Django, .NET Core, RESTful APIs, Microservices, MVC Frameworks, Docker, Kubernetes, AWS S3, Microsoft Azure, Jenkins, Git, Shell Scripting

Experience

Arctrade, Web Development Team · **Software Engineer (Full-time)**

Toronto, ON
Aug 2025 – Present

- Developed and managed the full-stack web infrastructure for company-wide workflow management systems, focusing on both frontend and backend development.
- Designed and implemented modular workflow templates to standardize complex business logic, allowing for consistent deployment across different departments.
- Engineered automated execution instances to process standardized workflows, ensuring reliable and repeatable execution of business tasks.
- Developed user-friendly web interfaces for monitoring and managing these templates and instances, providing real-time visibility into workflow status.
- Authored and maintained comprehensive test suites to ensure system stability during concurrent template creation and high-volume instance execution.
- Conducted routine maintenance of backend services and databases to support the workflow management platform's performance and scalability.

Quantum Management Services Limited, Technical Support Team · **Support Engineer (Full-time)**

Toronto, ON
Jul 2024 – Jul 2025

- Developed and managed a comprehensive CI/CD (Continuous Integration/Continuous Deployment) pipeline for internal development teams to automate software building, installation, and deployment.
- Automated end-to-end software testing within the pipeline, including recording results and performing historical comparisons to identify performance regressions.
- Created and maintained a centralized web dashboard to aggregate all test data, providing visualization of key performance metrics for the development group.
- Conducted detailed analysis of test data to identify system bottlenecks and provided performance optimization recommendations to senior developers.
- Maintained and configured physical and virtual hardware dedicated to automated testing, ensuring 24/7 availability of the testing environment.
- Provided direct technical support to developers, assisting with pipeline configuration and developing custom testing modules based on specific requirements.

C-MORE Lab, Robotics Team · **Software Engineer (Intern)**

Toronto, ON
May 2023 – Dec 2023

- Developed a specialized Python-based automation script to convert 2D map images into 3D simulation scenes within the Gazebo platform.

- Facilitated virtual robot testing environments by automating the generation of complex 3D scenarios from static 2D image data.
- Collaborated with the robotics researchers to validate the accuracy of 3D scene reconstruction and improved navigation script reliability through iterative testing.
- Maintained routine software tools used for lab data visualization and simulation configuration.

Intel Corp., Programmable Solutions Group (PSG) · Software Engineer (Intern)

Toronto, ON
May 2022 – Jul 2023

- Streamlined internal chip analysis web tools by combining PostgreSQL query tuning and Pandas processing, reducing end-to-end data processing latency by 85%.
- Implemented Django-powered dynamic query features, enabling 20+ engineers to customize and visualize chip data on the internal web platform.
- Managed a large-scale Quartus chip database (10M+ records) using PostgreSQL, ensuring data integrity and high-speed retrieval.
- Integrated the chip database into internal CI/CD regression testing pipelines to automate hardware performance verification.

Education

MEng University of Toronto, Computer Engineering

Sept 2024 – March 2026

- **Note:** This is a course-based Master's program. No thesis, no academic advisor, no publications, and no research funding involved.

BASc University of Toronto, Computer Engineering

Sept 2019 – June 2024

- **GPA:** 3.76/4.0
- **Graduate with Honours**
- **Dean's Honours List (6 Semesters)**